

Exploiting Metasploitable Report

This report documents the process of conducting a penetration testing exercise against the Metasploitable 2 virtual machine in a controlled lab environment. The objective was to identify vulnerabilities, exploit one of them successfully, and demonstrate access to the target system.

1) General Information

- **Target Domain Name:** Metasploitable
 - **Assessment Date:** 20/2/2026
 - **Analyst Name:** Mustafa Jbili
 - **Scope of Work:** Cybersecurity Analyst
 - **Objective:** exploit an Metesploitable machine with metasploit
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2) Methodology & Tools

- **Methodology:** identify vulnerabilities on metasploitable and attempt to exploit it and gain access
- **Tools Used:**
 - Nmap
 - Metasploit

3) Environment Setup

3.1 Lab Configuration

The penetration testing lab consisted of the following:

- Attacker machine: Kali Linux
 - Target machine: Metasploitable 2
 - Both machines were connected to the same virtual network.
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3.2 Network Verification

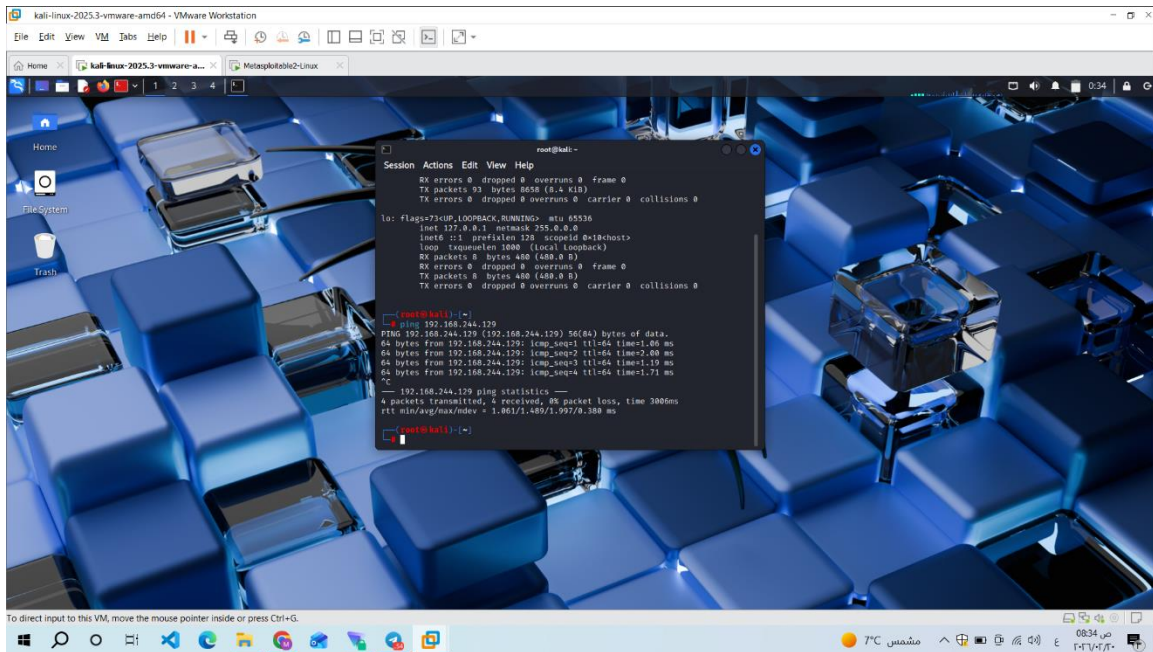
Connectivity between the attacker and target was verified using the ping command.

Target IP address:

192.168.244.129

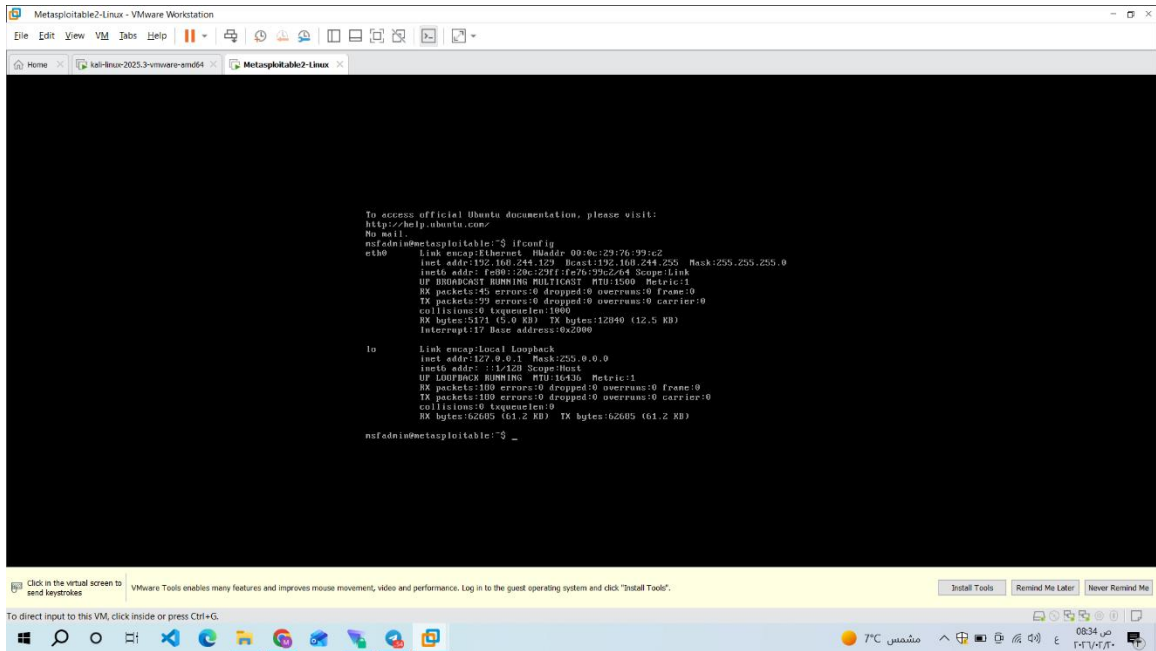
Ping test result:

Successful



The screenshot shows a Kali Linux desktop environment with a blue keyboard background. A terminal window is open, displaying the following output:

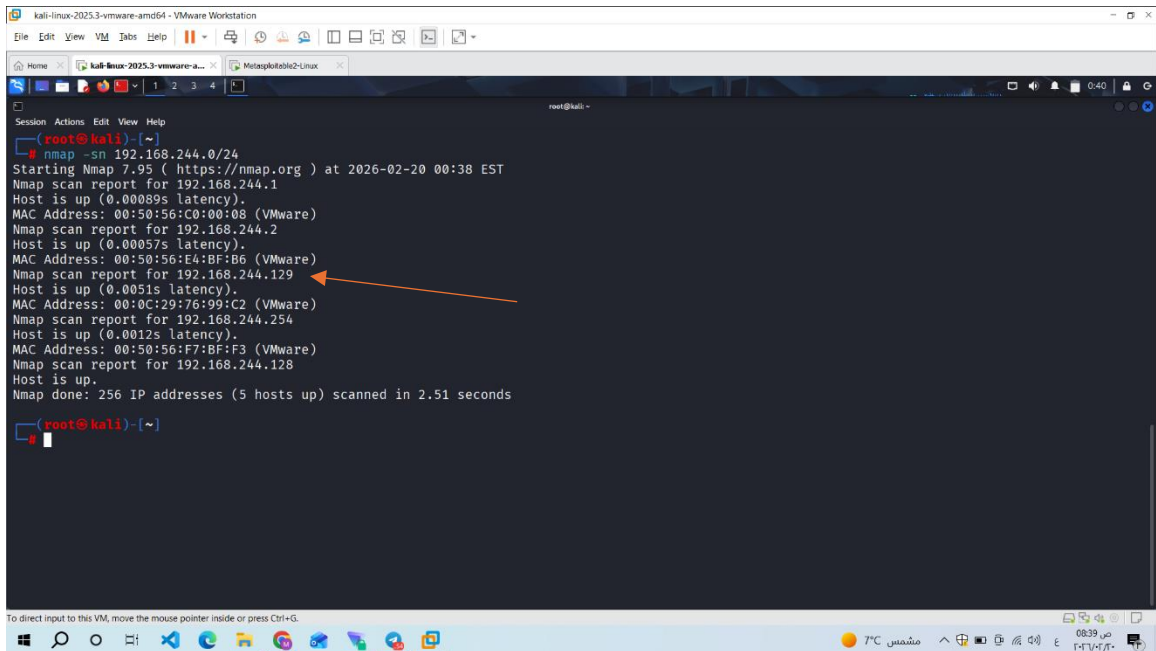
```
root@kali:~# ip netns exec ns1 ip netns exec ns2 ping -c 1 192.168.244.129
PING 192.168.244.129 (192.168.244.129) 56(64) bytes of data:
64 bytes from 192.168.244.129: icmp_seq=1 ttl=64 time=1.06 ms
64 bytes from 192.168.244.129: icmp_seq=2 ttl=64 time=1.00 ms
64 bytes from 192.168.244.129: icmp_seq=3 ttl=64 time=1.19 ms
64 bytes from 192.168.244.129: icmp_seq=4 ttl=64 time=1.71 ms
^C
--- 192.168.244.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 1.061/1.489/1.997/0.300 ms
```



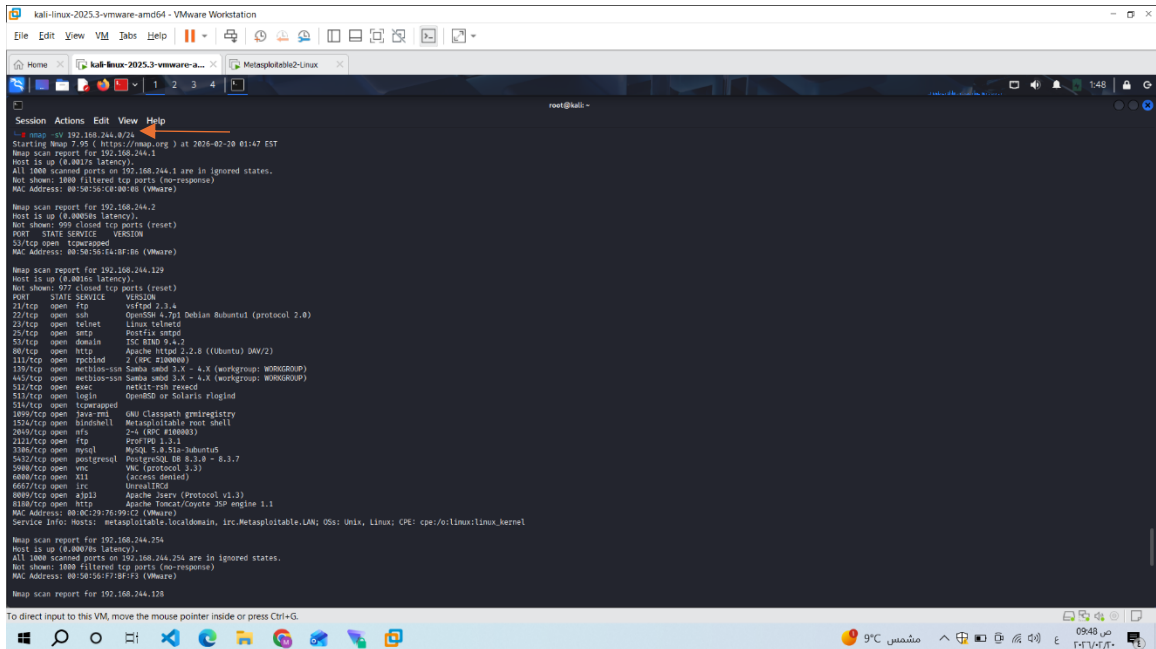
4) Vulnerability Scanning

4.1 Nmap Scan

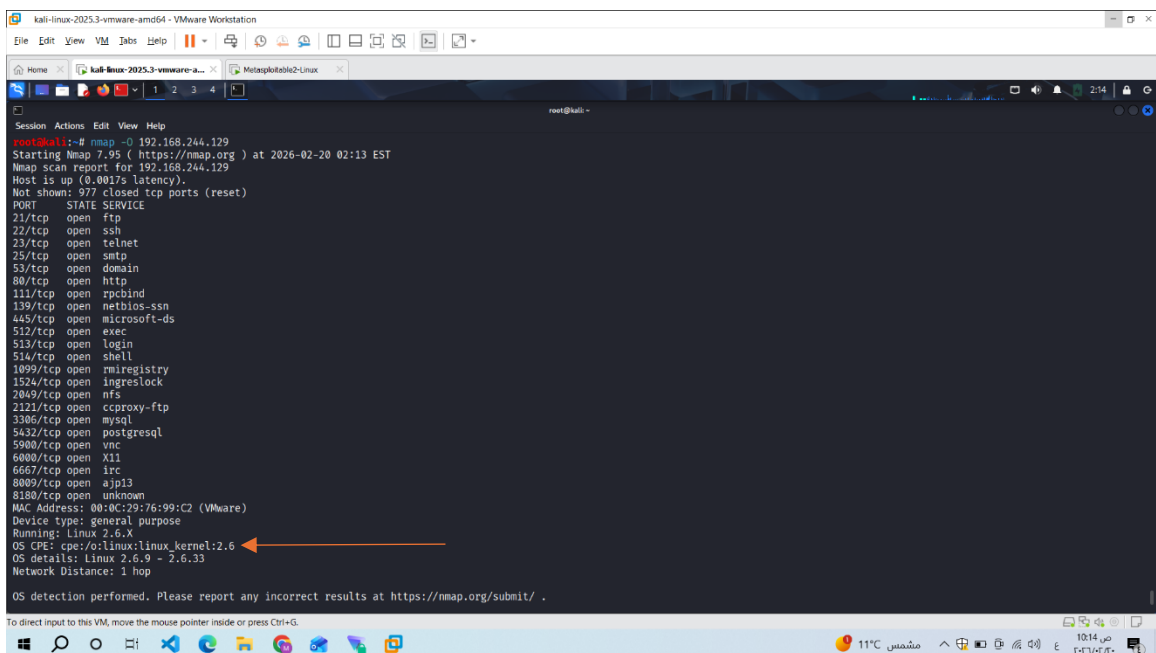
An initial scan was conducted to identify active devices on the target network.



Then..identify open ports and running services on the target



The operating system type and version used have been identified.



5) Vulnerability Identification

Based on the scan results, the following vulnerable service was identified:

Service name:Irc -6667

Reason for selection:This service is known to contain publicly documented vulnerabilities and exploit modules within Metasploit.

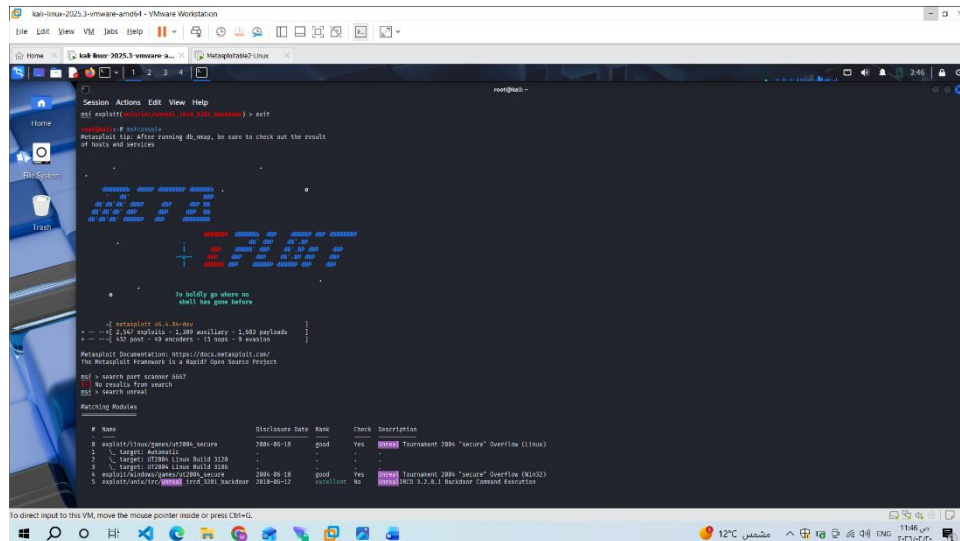
6) Exploitation

6.1 Metasploit Usage

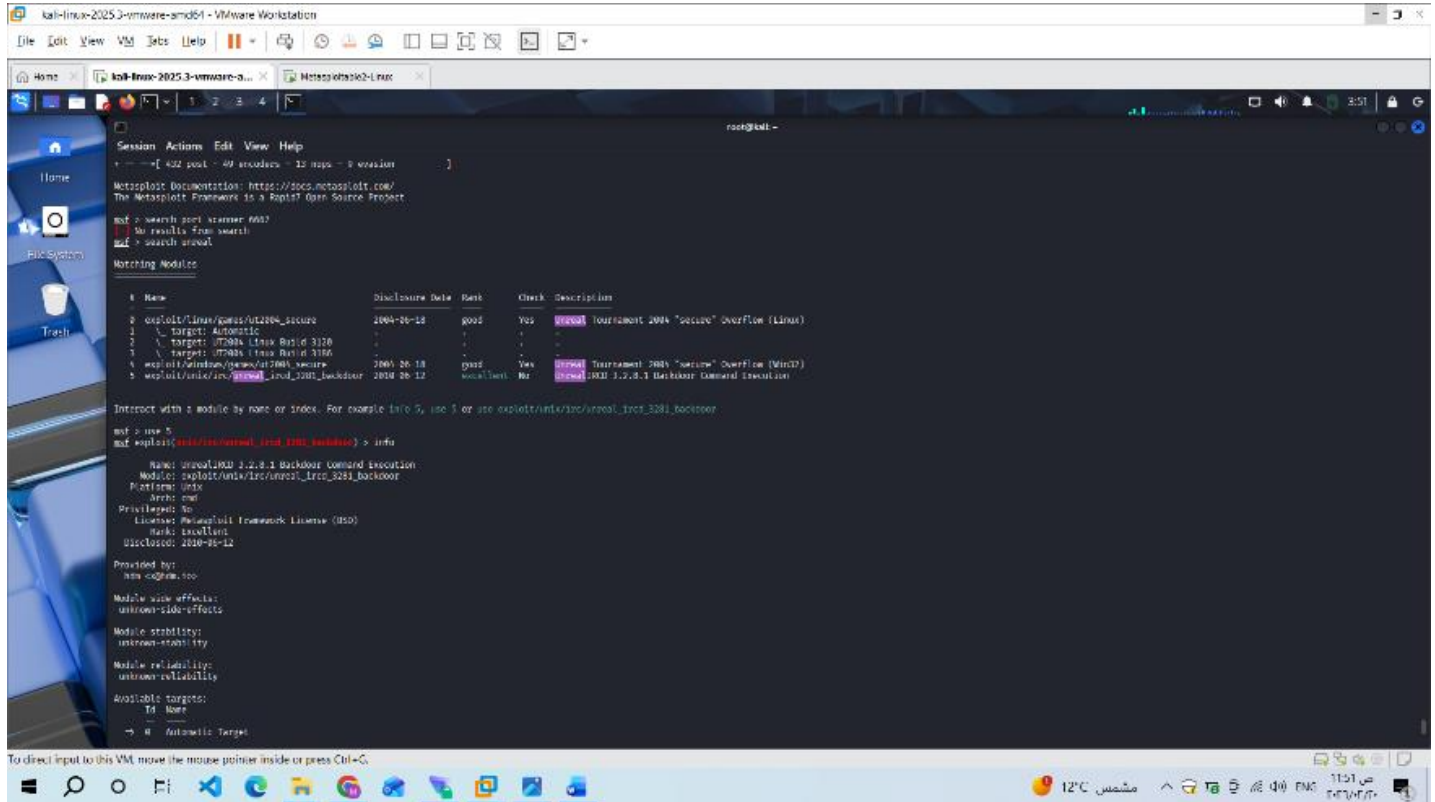
The **Metasploit Framework** was used to exploit the identified vulnerability.

Steps performed:

1.Started Metasploit console :



2. Searched for relevant exploit:



```
kali-linux-2025.3-vmware-ami64 - VMware Workstation
File Edit View VM Jobs Help
kali-linux-2025.3-vmware-ami64 x Metasploit-Meterpreter
root@kali:~#
Session Actions Edit View Help
+ --[ 432 post - 49 encoders - 12 nops - 0 version ]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid 0day Source Project

root> search post scanner 6667
No results from search
root> search usual

Matching Modules

#  Name                               Disclosure Date  Rank  Check  Description
#  ---                               -
0  exploit/linux/guest/ut2004_secure    2004-05-18      good  Yes    Tournament 2004 "secure" Overflow (Linux)
1  \_ target: Automatic                -              -      -      -
2  \_ target: UT2004 (Linux Build 3120) -              -      -      -
3  \_ target: UT2004 (Linux Build 3166) -              -      -      -
4  exploit/windows/guest/ut2004_secure  2004-06-18      good  Yes    Tournament 2004 "secure" Overflow (Win32)
5  exploit/linux/guest/ut2004_secure    2004-06-12      good  Yes    Tournament 2004 "secure" Overflow (Linux)

Interact with a module by name or index. For example info 5, use 1 or use exploit/linux/guest/ut2004_secure

root> use 5
root> exploit(windows/guest/ut2004_secure) > info
Name: ut2004 2.2.8.1 Backdoor Command Execution
Module: exploit/linux/guest/ut2004_secure
Platform: linux
Arch: x86
Privileged: No
License: Metasploit Framework License (BSD)
Rank: Excellent
Disclosure: 2004-05-18

Provided by:
hms cd@me.too

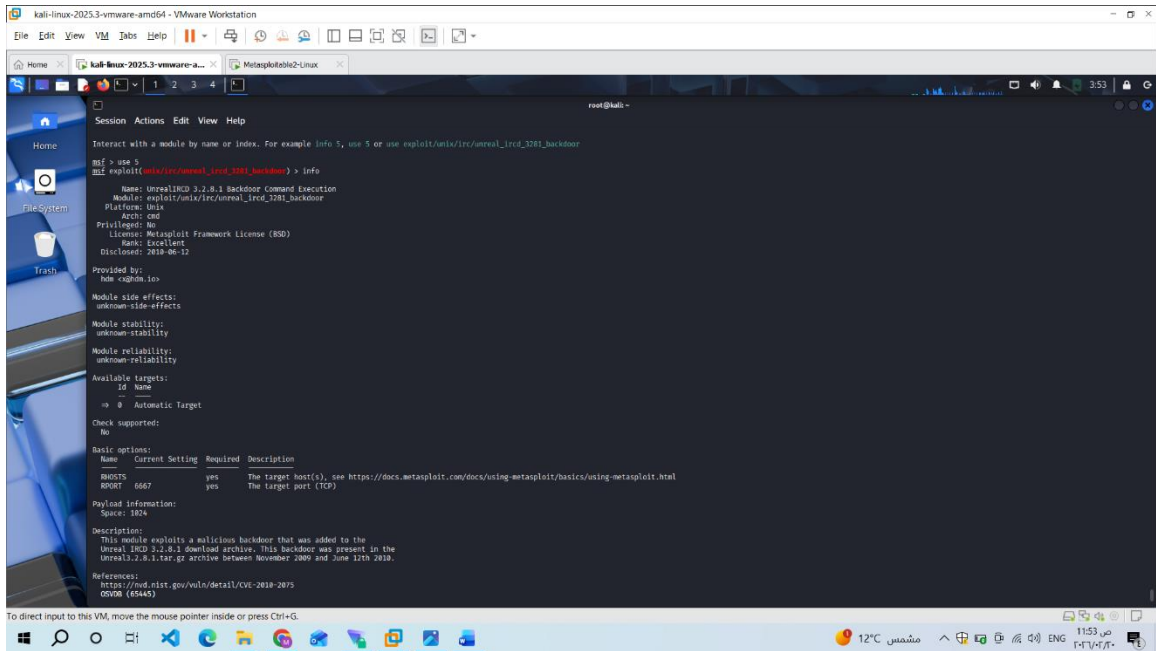
Module side effects:
unknown-side-effects

Module stability:
unknown-stability

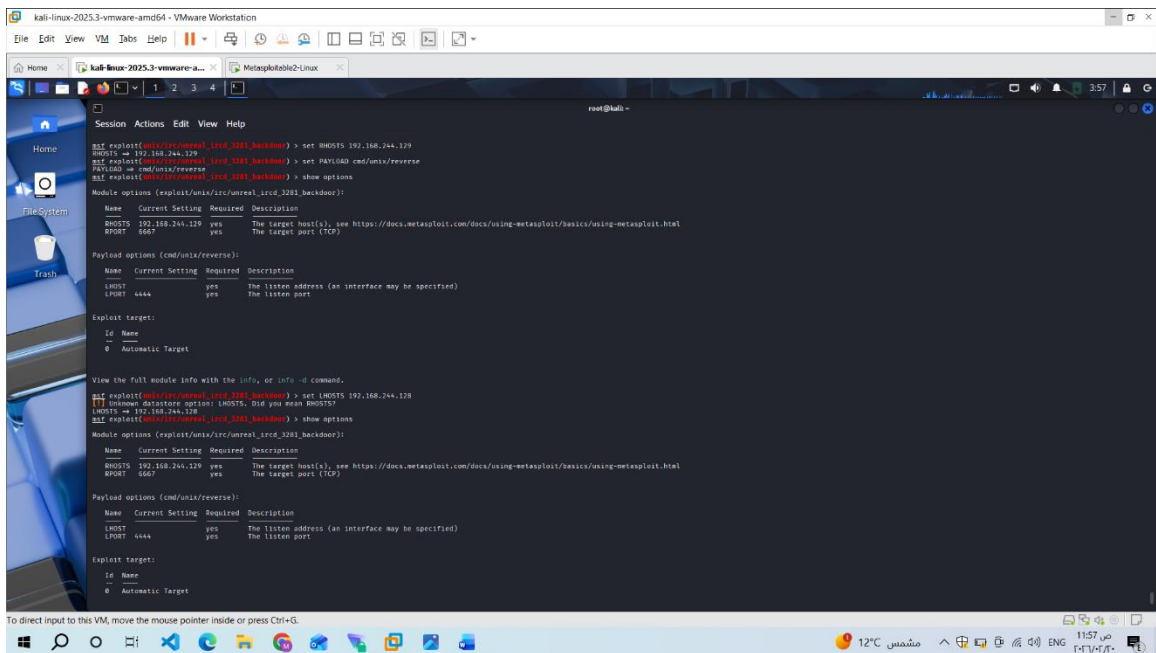
Module reliability:
unknown-reliability

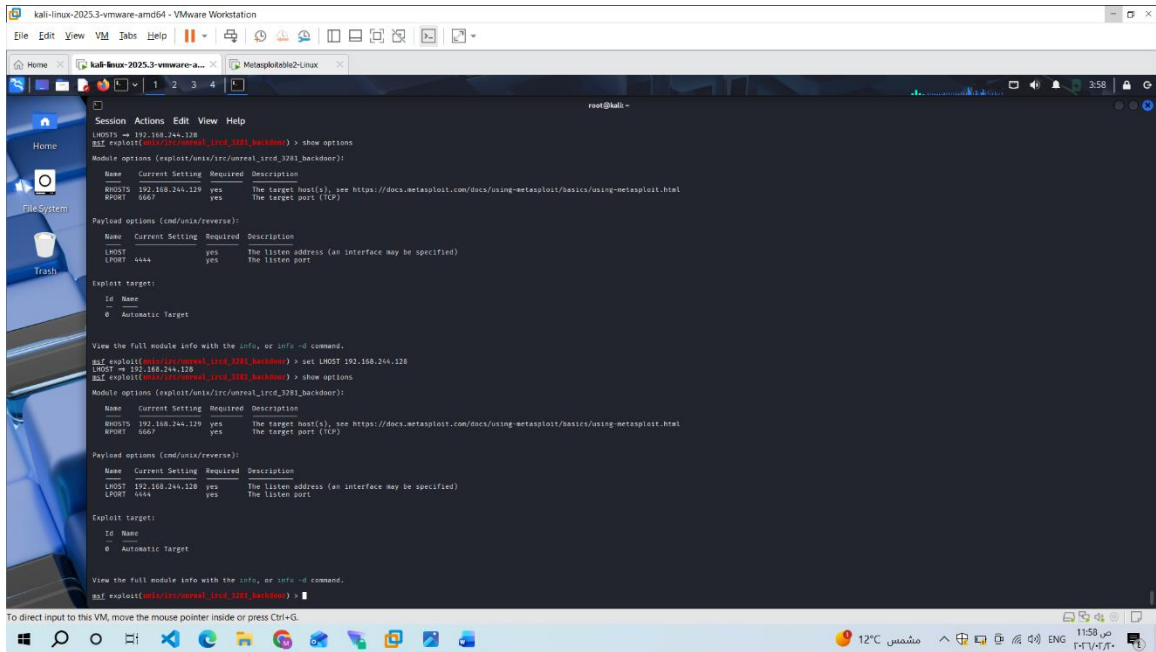
Available targets:
#  Name
#  ---
0  Automatic Target
```

3. Selected exploit module:

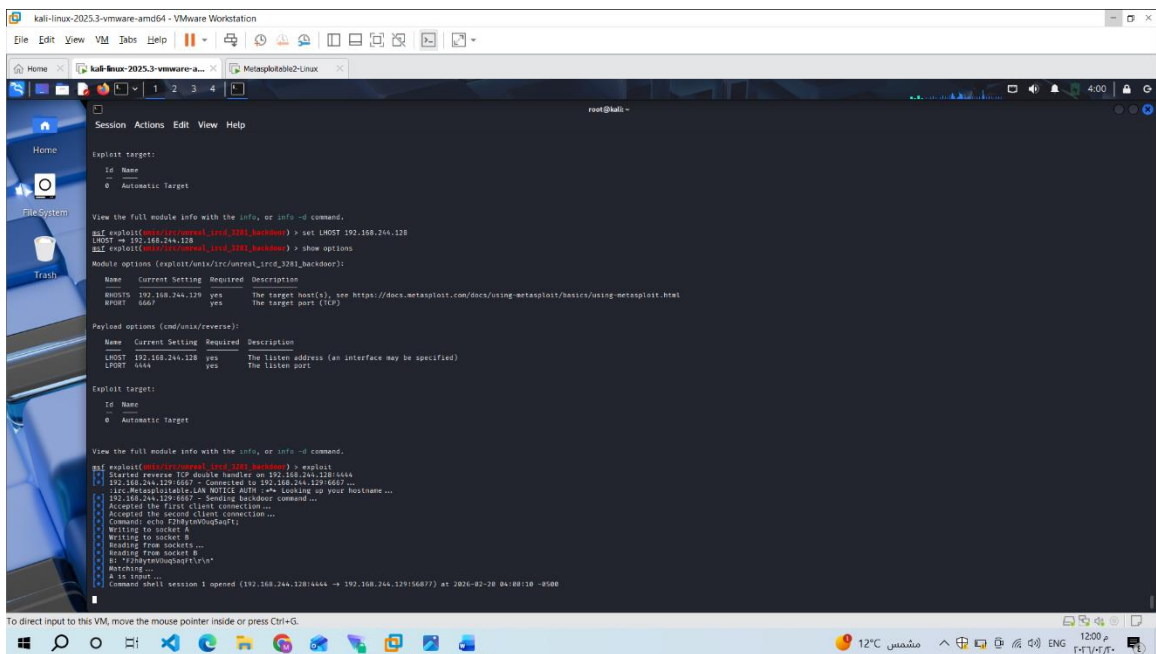


4. Configured required options:





5.Executed the exploit:



7) Post-Exploitation

After gaining access to the system, the following commands were executed to verify control over the compromised host:

[illegible]